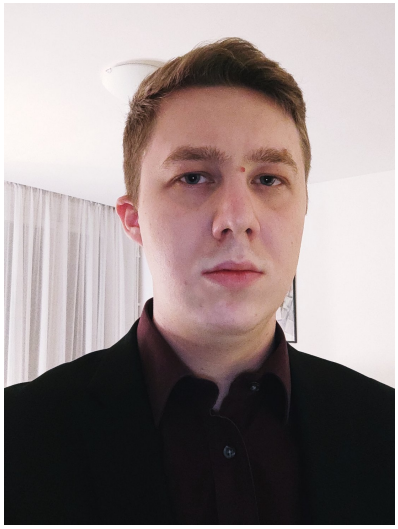


Danijel G. Aleksić, PhD
Mathematician

Born on September 25, 1998, in Zvornik. Enrolled at the Faculty of Mathematics, University of Belgrade in 2017, within the Mathematics study program, module Statistics, Actuarial and Financial Mathematics, where he graduated in 2021 with a GPA of 9.47. Obtained a Master's degree one year later with a GPA of 8.75 within the same study program and module. Defended his doctoral dissertation in the field of U- and V-statistics applications for incomplete data in 2026 at the Faculty of Mathematics.

His primary research interests include the theoretical analysis of missing data mechanisms, as well as the application of U- and V-statistics to various types of incomplete data.

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Software

- **LaTeX**
- **R**
- **GNU/Linux**
- **Python**
- **MS Excel**

Languages

- **Serbian**
- **English**
- **Russian**

Education

PhD in Mathematics — **X 2022 – IV 2026**
University of Belgrade, Faculty of Mathematics
Doctoral Dissertation:
U- and V- statistics for incomplete data and their application to model specification testing
Advisor: Prof. Bojana Milošević, PhD

Master's Degree in Mathematics — **X 2021 – IX 2022**
University of Belgrade, Faculty of Mathematics
Module: Statistics, actuarial and financial mathematics
GPA: 8.75 (out of 10).
Thesis:
The problem of missing data: Impact on statistical inference (in Serbian)
Advisor: prof. Bojana Milošević, PhD
Laureate of the *Annual Award of the Mathematical Institute of the Serbian Academy of Sciences and Arts in the field of mathematics and mechanics for M.Sc. students*

B.Sc. in Mathematics — **IX 2017 – IX 2021**
University of Belgrade, Faculty of Mathematics
Module: Statistics, actuarial and financial mathematics
GPA: 9.47 (out of 10).

***Vuk Karadžić* Grammar School, Loznica** — **IX 2013 – V 2017**
Science and Mathematics Track
GPA: 5.0 (out of 5). Laureate of *Vuk Karadžić Diploma*.

***Vuk Karadžić* Elementary School, Roćević, Zvornik** — **IX 2004 – V 2013**
GPA: 4.93 (out of 5).

Work Experience

Teaching Assistant — **XI 2023 –**
University of Belgrade, Faculty of Organizational Sciences
Department of Mathematics

Courses:

- **Mathematics 1** (*Algebraic structures, basics of linear algebra, analytic geometry in space, sequences of real numbers, differential calculus of real functions of one real variable*)
- **Mathematics 2** (*Continuity, limit values and differential calculus of multivariate functions, integration of real-valued univariate and multivariate functions*)

Part-time Teaching Assistant at the Faculty of Mathematics, Department of Probability and Statistics, during the first semester of the school year of 2023/24. Courses:

- **Introduction to financial mathematics** (Fixed interest rates, variable interest rates, loans, bonds)

Part-time mathematics teacher at Mathematical Grammar School, Belgrade, during the school year of 2024/25. Courses:

- **Mathematical Analysis and Algebra**, 4th grade (Applications of derivatives, indefinite and definite integral, introduction to differential equations)

Teaching Associate — **XI 2022 – X 2023**
University of Belgrade, Faculty of Organizational Sciences
Department of Mathematics

Courses:

- **Mathematics 1**
- **Mathematics 2**

Part-time Teaching Associate at the Faculty of Mathematics, Department of Probability and Statistics, in the second semester of the school year of 2022/23. Courses:

- **Probability and Statistics B** (*Characteristic function, convergence of sequences of random variables, point and interval estimates of parameters, hypothesis testing*)
- **Information theory** (*Shannon's entropy, information, coding theory, linear codes*)

Teaching Associate X 2021 – X 2022
University of Belgrade, Faculty of Mathematics
Department of Probability and Statistics

Courses:

- **Statistical software 3** (*Introduction to the statistical and machine learning using the programming language R*)
- **Introduction to the sample theory** (*Sampling with and without replacement, stratified sample, systematic sample, cluster sample*)
- **Information theory**

Part-time Teaching Associate at the Faculty of Organizational Sciences, Department of Mathematics, in the second semester of the school year of 2021/22. Courses:

- **Mathematics 2**

Publications

Theses

1. **Aleksić, D.G.** (2026). U- and V- statistics for incomplete data and their application to model specification testing (Doctoral dissertation). *University of Belgrade, Faculty of Mathematics, Belgrade, Serbia.* `http://elibrary.matf.bg.ac.rs/bitstream/handle/123456789/5781/DanijelAleksicPhDThesis.pdf?sequence=1`
2. **Aleksić, D.** (2022). The problem of missing data: Impact on statistical inference (Master's thesis) ***(in Serbian)***. *University of Belgrade, Faculty of Mathematics, Belgrade, Serbia.* `http://elibrary.matf.bg.ac.rs/bitstream/handle/123456789/5465/DanijelAleksicMaster.pdf`

Journal Articles

1. **Aleksić D.G.**, & Milošević, B. (2024). To impute or not? Testing multivariate normality on incomplete dataset: revisiting the BHEP test. *Journal of Applied Statistics* 52(9), 1742–1759. `https://doi.org/10.1080/02664763.2024.2438798`
2. **Aleksić, D.** (2024). A novel test of missing completely at random: U-statistics-based approach. *Statistics*, 58(4), p. 1004–1023. `https://doi.org/10.1080/02331888.2024.2386361`
3. **Aleksić D.**, Cuparić M., & Milošević, B. (2023). Non-degenerate *U*-statistics for data missing completely at random with application to testing independence. *Stat* 12(1), e634. `https://doi.org/10.1002/sta4.634`

Preprints/Manuscripts Under Review

1. **Aleksić, D.G.**, Džamić, D., Milošević, B., & Obradović, M. (2026). Extending an Order-Statistic Characterization of the Exponential Distribution. *arXiv preprint arXiv:2607.13302*
2. **Aleksić, D.G.** (2026). Testing MCAR via covariances: Extending the U-statistic framework to partially observed variables. *arXiv preprint arXiv:2501.05596*
3. **Aleksić, D.G.**, & Milošević B. (2025). Two-sample testing with missing data via energy distance: Weighting and imputation approaches. *arXiv preprint arXiv:2508.11421*

Talks

Conferences—Invited Talks

1. **Aleksić, D.** (2025, July 21–25). *Testing goodness-of-fit, independence and MCAR on incomplete dataset utilising the properties of non-degenerate and weakly-degenerate U- and V- statistics.* 24th European Young Statisticians Meeting, Collegio Carlo Alberto, Torino, Italy.
   https://sites.google.com/view/eysmtorino2025/book-of-abstracts, page 15.
2. **Aleksić, D.** (2023, December 13–15). *A new approach to Little's MCAR test.* CMStatistics, HTW Berlin, University of Applied Sciences, Berlin, Germany.
   https://www.cmstatistics.org/CMStatistics2023/docs/BoA.pdf?20231128014621, page 43.

Conferences—Other Talks

1. **Aleksić, D.G.** (2025, December 12–13). *Flexible MCAR testing for missing data with correlated response indicators.* XV Symposium "Mathematics and Applications", University of Belgrade, Faculty of Mathematics, Belgrade, Serbia.
   https://simpozijum.matf.bg.ac.rs/s2025/AleksicDanijel.pdf
2. **Aleksić, D.**, Cuparić, M., & Milošević, B. (2025, October 2–5). *Asymptotic results for some U- and V-statistics for model specification testing under missing data.* 3rd Joint Mathematical Meeting of Serbia and Montenegro, Petrovac, Montenegro. (presenting author: Aleksić, D.)
3. **Aleksić, D.**, & Milošević, B. (2025, August 28–31). *Homogeneity testing in the presence of missing data.* Workshop on Goodness-of-fit, Change-point and Related Problems, Faculty of Mathematics and Physics, Charles University, Prague, Czechia. (presenting author: Milošević, B.)
   https://gofcp.karlin.mff.cuni.cz/GOFCP_book_of_abstracts.pdf, page 21.
4. Milošević, B., & **Aleksić, D.** (2025, July 8–9). *Energy-distance-based two-sample testing in the presence of incomplete data.* HiTEc & Codes, Cyprus University of Technology, Limassol, Cyprus. (presenting author: Milošević, B.)
   https://www.cmstatistics.org/hiteccodes2025/fullprogramme.php
5. **Aleksić, D.**, & Milošević, B. (2024, June 25–29). *Addressing missing data challenges: A multivariate goodness-of-fit testing perspective.* 6th International Symposium on Nonparametric Statistics, Braga, Portugal. (presenting author: Milošević, B.)
   https://w3.math.uminho.pt/ISNPS2024/img/ISNPS%202024_PROGRAMME.pdf, page 88.
6. **Aleksić, D.**, & Milošević, B. (2024, June 19–22). *To impute or not to? A multivariate goodness-of-fit testing perspective.* 15th Serbian Mathematical Congress, University of Belgrade, Faculty of Mathematics, Belgrade, Serbia. (presenting author: Aleksić, D.)
   https://smak15.matf.bg.ac.rs/smak15/AleksicMilosevic15Smak.pdf
7. Cuparić, M., **Aleksić, D.**, & Milošević, B. (2024, March 23–24). *Testing independence in the presence of data missing completely at random.* The Spring HiTEc meeting and the Complex Data in Econometrics and Statistics Workshop, Cyprus University of Technology, Limassol, Cyprus. (presenting author: Cuparić, M.)
   https://www.cmstatistics.org/hiteccodes2024/fullprogramme.php
8. **Aleksić, D.** (2023, December 1–2). *U-statistics-based approach for testing the MCAR assumption.* XIII Symposium "Mathematics and Applications", University of Belgrade, Faculty of Mathematics, Belgrade, Serbia.
   http://simpozijum.math.rs/s2023/DanijelAleksicApstrakt.pdf
9. **Aleksić, D.**, Cuparić, M., & Milošević, B. (2023, September 29–30). *On the asymptotic properties of non-degenerate U-statistics in the presence of MCAR data.* Statistical Modeling with Applications, Casa Academiei Române, Bucharest, Romania. (presenting author: Aleksić, D.)
   https://statmod2023.sciencesconf.org/data/pages/BoA_StatMod2023_WithPoster_2.pdf, page 3.

10. **Aleksić, D.**, & Milošević, B. (2023, January 26–28). *Missing data: the impact on multivariate goodness-of-fit tests.* 2nd Joint Mathematical Meeting of Serbia and Montenegro, Mathematical Institute of the Serbian Academy of Sciences and Arts, Belgrade, Serbia. (presenting author: Aleksić, D.) https://konferencije.matf.bg.ac.rs/smscg2023/wp-content/uploads/2023/01/smscg2023-knjiga-apstrakta-1-1.pdf, page 1.

11. Milošević, B., & **Aleksić, D.** (2022, December 17–19). *Testing multivariate normality in the presence of missing data.* CMStatistics, King's College, London, United Kingdom. (presenting author: Milošević, B.) http://www.cmstatistics.org/CMStatistics2022/docs/BoACFECMStatistics2022.pdf, page 117.

12. **Aleksić, D.**, & Milošević, B. (2021, December 2–3). *The impact of imputation methods on multivariate normality tests.* XI Symposium "Mathematics and Applications", University of Belgrade, Faculty of Mathematics, Belgrade, Serbia. (presenting author: Aleksić, D.) ***(in Serbian)*** https://alas.matf.bg.ac.rs/~konferencija/KNJIGA_APSTRAKATA_2021.pdf, page 1.

13. **Aleksić, D.** (2021, September 20–23) Imputation of missing data in univariate time series. *XLVIII nternational Symposium on Operational Research, Banja Koviljača, Serbia.* ***(in Serbian)*** http://symopis2021.matf.bg.ac.rs/download/Zbornik-SYM-OP-IS2021.pdf, page 456.

14. **Aleksić, D.** (2019, December 6–7). *Relationship between coin thickness and the probability of landing on its edge.* X Symposium "Mathematics and Applications", University of Belgrade, Faculty of Mathematics, Belgrade, Serbia. ***(in Serbian)*** https://alas.matf.bg.ac.rs/~konferencija/KNJIGA_APSTRAKATA_2019.pdf, page 37.

Seminars

1. **Aleksić, D.** (15 April 2025). *A new test for missing completely at random and its importance in the context of goodness-of-fit tests.* University of Belgrade, Faculty of Mathematics, Seminar of the Department of Probability and Statistics, Belgrade, Serbia. ***(in Serbian)***

2. **Aleksić, D.** (28 March 2025). *Differential privacy: the challenge of its preservation in the case of an incomplete sample.* University of Belgrade, Faculty of Mathematics, Seminar of the Department of Probability and Statistics, Belgrade, Serbia. ***(in Serbian)***

Review Activity

D.G. Aleksić has been a referee for manuscripts submitted to the following journals: *Annals of Statistics*, *Journal of Statistical Computation and Simulation*, *Journal of Applied Statistics*, *Thermal Science*, *BMC Medical Research Methodology*, *International Journal of Data Science and Analytics*, *Journal of Multivariate Analysis*.

Mentorship and Student Research Development

Committee Memberships

D. G. Aleksić served as a committee member for the defense of the following Bachelor's (Diploma) theses at the Faculty of Organizational Sciences, University of Belgrade:

1. **Mihailo Todorović** (28 October 2025). Application of Voronoi Diagrams in Service Facility Location Planning. *University of Belgrade, Faculty of Organizational Sciences.* Mentor: Prof. Milica Stojanović, PhD.

2. **Maja Branković** (29 May 2024). The Golden Ratio in Music. *University of Belgrade, Faculty of Organizational Sciences.* Mentor: Prof. Vesna Todorčević, PhD.

Conference Committee Membership

- Member of the International Organizing and Program Committee of the conference *25th European Young Statisticians Meeting*, July 7–10, 2026, Vilnius, Lithuania.